

Power BI Basics Assignment

Question 1: What is Power BI and why is it used in businesses?

Answer:

Power BI is a business analytics and data visualization tool developed by [Microsoft](#). It helps users collect, analyze, and visualize data from different sources such as Excel files, databases, cloud services, and websites.

Businesses use Power BI because it allows them to:

1. Create interactive dashboards and reports.
2. Analyze large amounts of data quickly.
3. Make better business decisions using real-time insights.
4. Share reports easily with team members and management.
5. Track performance through charts, graphs, and KPIs (Key Performance Indicators).

Power BI is widely used in finance, sales, marketing, healthcare, and many other industries for data-driven decision-making.

Question 2: Name and explain the three main components of Power BI.

Answer:

The three main components of Power BI are:

1. Power BI Desktop

Power BI Desktop is a Windows application used to create reports and dashboards. It allows users to connect to data sources, clean and transform data, and build visualizations. It is mainly used by report developers and analysts.

2. Power BI Service

Power BI Service is a cloud-based online platform where reports and dashboards are published and shared. Users can access reports through a web browser and collaborate with others in the organization.

3. Power BI Mobile

Power BI Mobile is a mobile application available for smartphones and tablets. It allows users to view dashboards and reports anytime and anywhere. This helps managers and employees monitor business performance on the go.

Question 3: Explain the Power BI workflow.

Answer:

The workflow of Power BI involves the following steps:

1. **Connect to Data**
Power BI connects to different data sources such as Excel, SQL databases, cloud services, and web data.
 2. **Transform and Clean Data**
Using Power Query, the data is cleaned and transformed by removing errors, duplicates, and unnecessary information.
 3. **Create Data Model**
Relationships are created between different tables to organize the data properly for analysis.
 4. **Build Visualizations and Reports**
Charts, graphs, tables, and dashboards are created to represent the data visually.
 5. **Publish Reports**
Reports are published to the Power BI Service for sharing and collaboration.
 6. **Share and Access Reports**
Users can access reports through web browsers or mobile applications and make informed business decisions.
-

Question 4: List any four data cleaning tasks that can be performed in Power Query.

Answer:

Four common data cleaning tasks performed in Power BI Power Query are:

1. **Removing Duplicate Rows**
Duplicate records can be removed to maintain accurate data.
2. **Handling Missing Values**
Blank or null values can be replaced, removed, or filled with suitable values.
3. **Changing Data Types**
Data types such as text, number, or date can be corrected for proper analysis.
4. **Filtering Unwanted Data**
Unnecessary rows or columns can be filtered or removed to keep only useful information.